

Back-to-School Crossword Solutions

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1 6	2 3
3 5	3

ONE ACROSS

Since Justin and Kyle complete $\frac{1}{900}$ of the project per day,

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{900}.$$

Rearranging the equation gives $xy = 900x + 900y$, so $(x - 900)(y - 900) = 900^2$.

The original equation implies $x > 900$ and $y > 900$. Thus, each solution corresponds to a factorization of 900^2 into two positive factors.

Since $900 = 2^2 \cdot 3^2 \cdot 5^2$, we have $900^2 = 2^4 \cdot 3^4 \cdot 5^4$, which has $(4 + 1)^3 = 125$ positive divisors.

The divisors pair as d and $\frac{900^2}{d}$. Because 900^2 is a perfect square, one pair consists of the repeated divisor 900. Therefore, the number of pairs with $x \leq y$ is $\frac{125+1}{2} = \boxed{63}$.

ONE DOWN

The semiperimeter of the triangle is $s = \frac{13+14+15}{2} = 21$.

By Heron's formula, its area is $K = \sqrt{21(21-13)(21-14)(21-15)} = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$.

Hence, the inradius is $r = \frac{K}{s} = \frac{84}{21} = 4$, and the circumradius is $R = \frac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \frac{65}{8}$.

Euler's formula gives $OI^2 = R(R - 2r) = \frac{65}{8} \left(\frac{65}{8} - 8 \right) = \frac{65}{64}$.

Therefore, $64OI^2 = \boxed{65}$.

TWO DOWN

We first count all proper colorings up to rotation, without requiring all three colors to be used.

The number of proper colorings of a cycle of length n using three colors is $2^n + 2(-1)^n$.

The identity rotation therefore fixes $2^8 + 2 = 258$ colorings.

Each rotation by one, three, five, or seven vertices fixes no proper colorings. A rotation by two or six vertices fixes $3 \cdot 2 = 6$ proper colorings, while a rotation by four vertices fixes $2^4 + 2 = 18$ proper colorings.

By Burnside's lemma, the number of proper colorings up to rotation is $\frac{258+2 \cdot 6+18}{8} = 36$.

It remains to remove the colorings that use exactly two colors. There are $\binom{3}{2} = 3$ choices for the two colors. For each choice, the vertices must alternate between the two colors, and the two possible alternating arrangements are equivalent under rotation. Thus, there is one equivalence class for each pair of colors.

Consequently, the number of colorings that use all three colors is $36 - 3 = \boxed{33}$.

THREE ACROSS

By Vieta's formulas, $a + b + c = 0$, $ab + bc + ca = -3$, and $abc = -1$.

Therefore, $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = 6$, and $a^2b^2 + b^2c^2 + c^2a^2 = (ab + bc + ca)^2 - 2abc(a + b + c) = 9$.

Using $(u + v)(v + w)(w + u) = (u + v + w)(uv + vw + wu) - uvw$ with $u = a^2$, $v = b^2$, and $w = c^2$, we obtain $(a^2 + b^2)(b^2 + c^2)(c^2 + a^2) = 6 \cdot 9 - (abc)^2 = 54 - 1 = \boxed{53}$.